zPSC Calibration and Test Manual

Michael Capotosto

May 2026

Contents

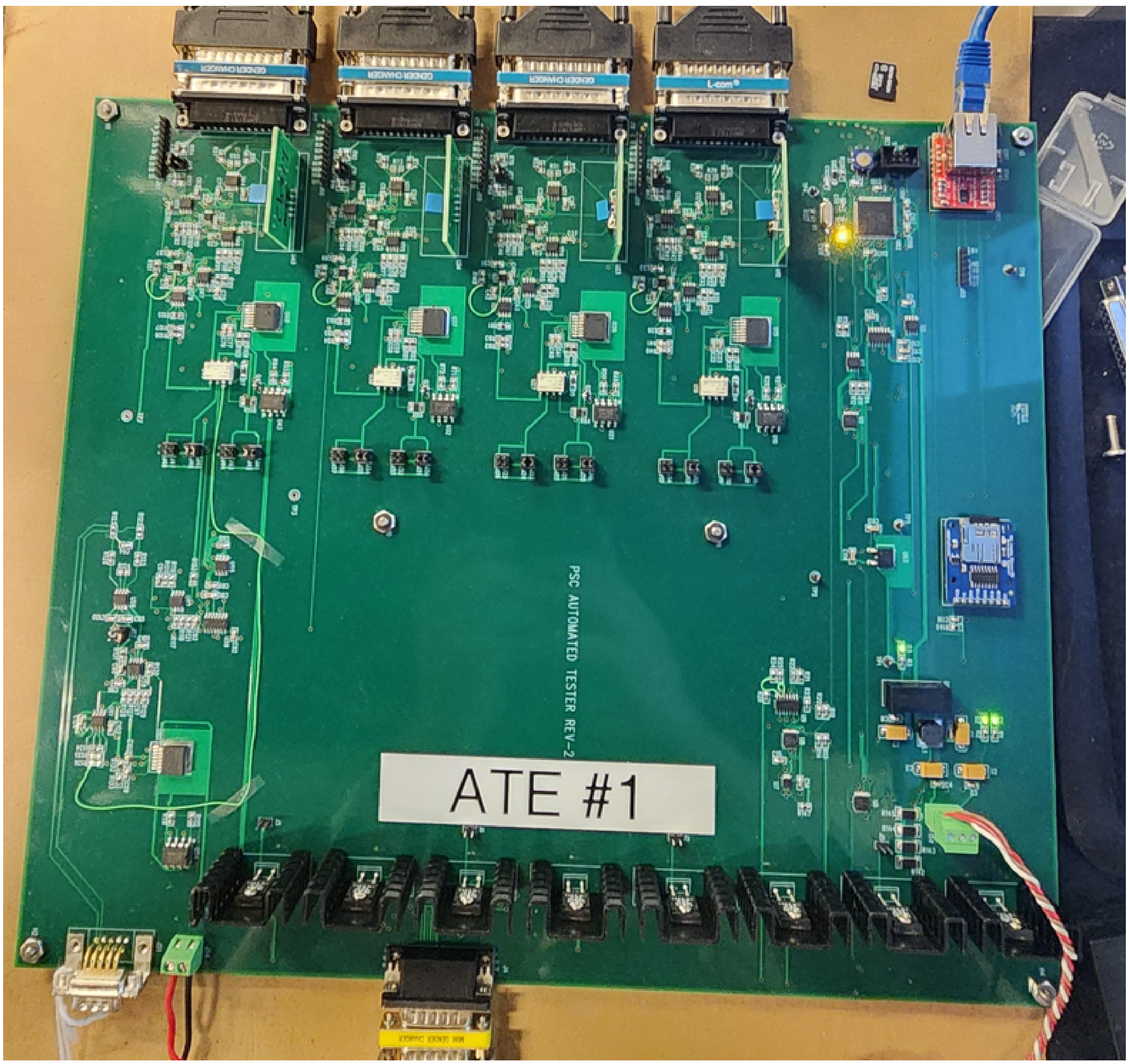
Chapter 1

Introduction

Chapter 2

ATE Connections

2.1 Overview



On initial setup, there are a few fixed connections that need to be made to the ATE:

- Current Shunt: Connect this to a resistance standard for calibrations, or to a wire loop if not being used for current measurement
- $\pm 15V$ Supply: Connect this to a split supply, draws less than 400mA under full load
- Network: Connect to the local test network
- SD Card must be configured once, but does not require any regular changes
- Jumpers: Install jumpers in J8-9, J12-13, J19-20, JJ23-24, J30-31, J34-35, J41-42, and J45-J46 for normal operations.

For every device under test:

- Channel 1, 2, 3, and 4 Power Supply Cables: These cables connect to each of the micro-D Power Supply I/O connectors on the rear of the PSC
- Tuning Boards: Tuning boards must be installed to match the loads being tested.
- DCCT Cable must be connected for every PSC. There are two cables - one for 4-Channel units, and another for 2-Channel units.

2.2 Connections

2.2.1 At the ATE

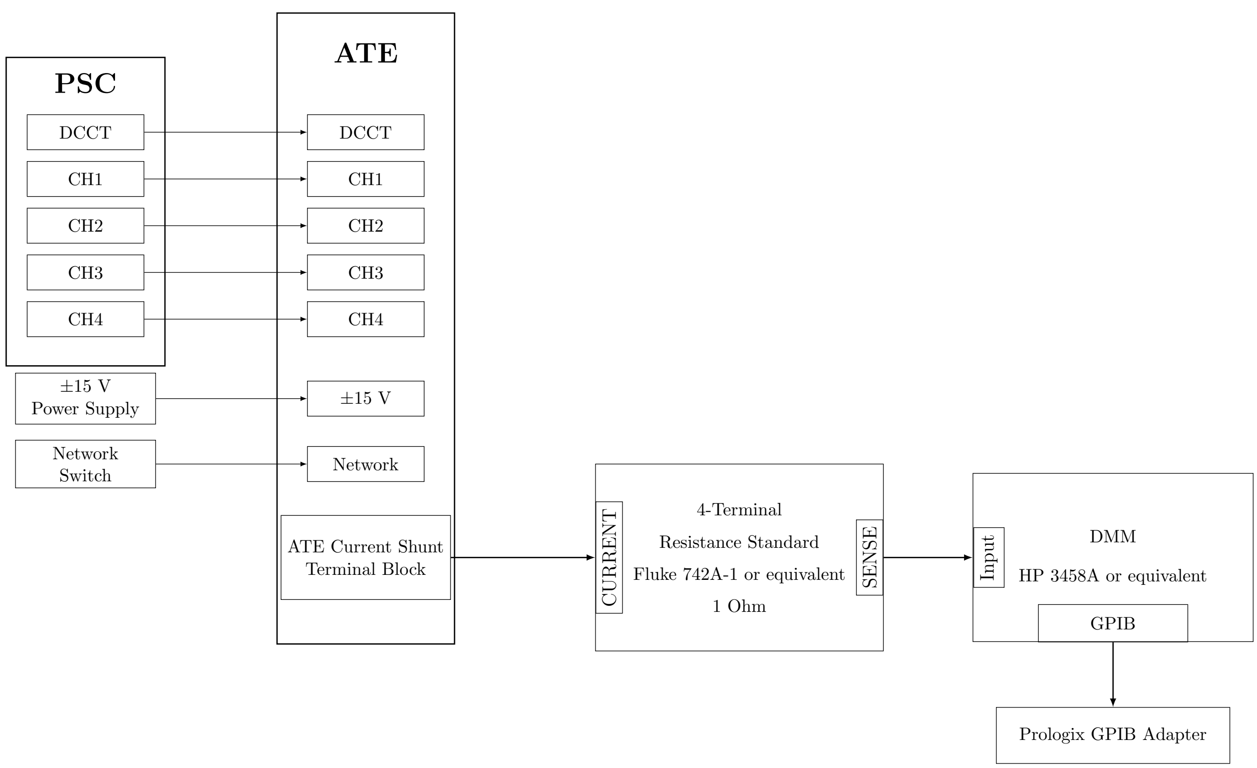


Figure 2.1: ATE Connection Block Diagram

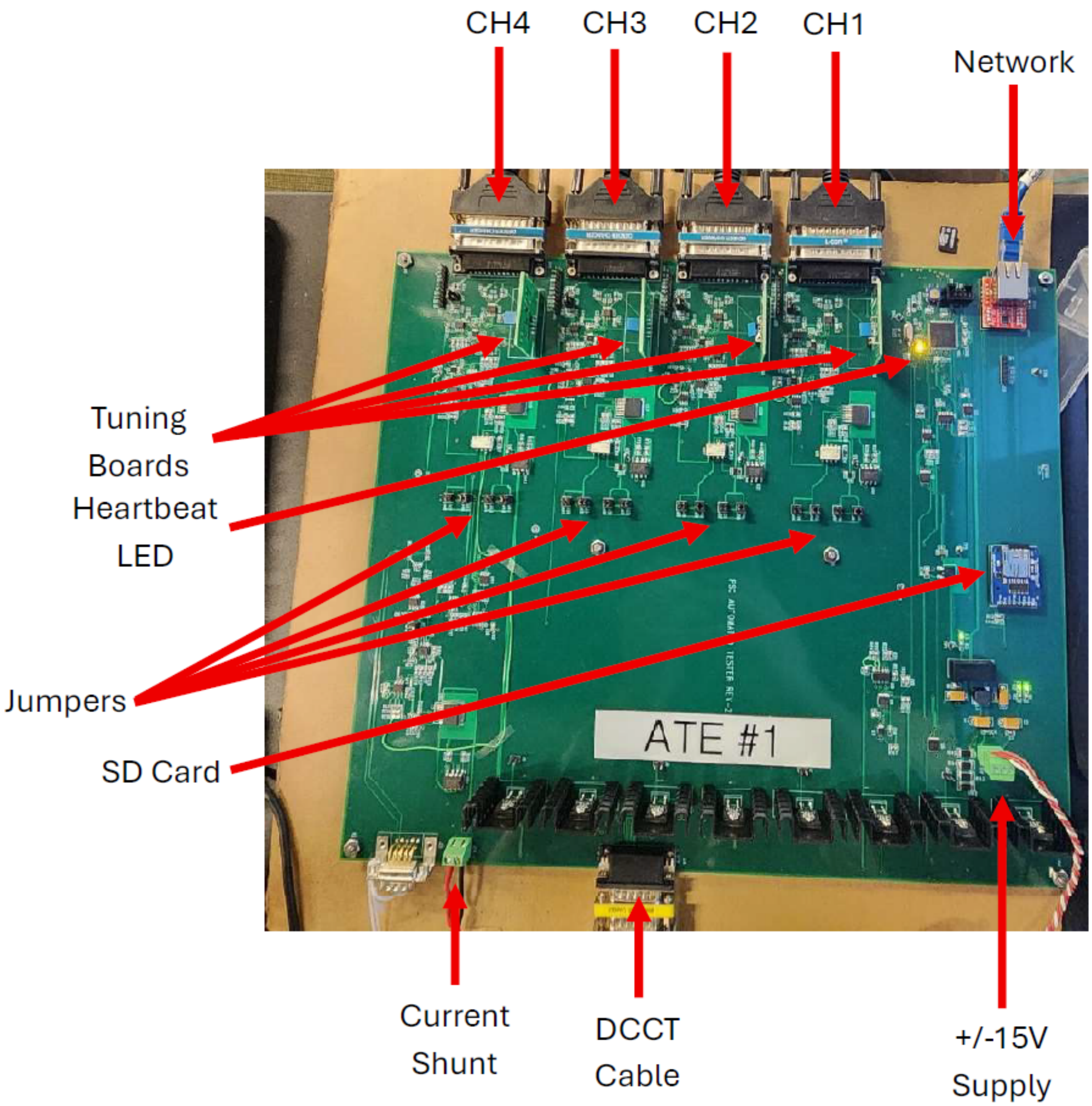


Figure 2.2: Annotated ATE Board

2.2.2 Current Measurement Loop/DMM Connection

The DMM is connected to the workstation via a GPIB adapter from Prologix. These are available in both USB and Ethernet forms.

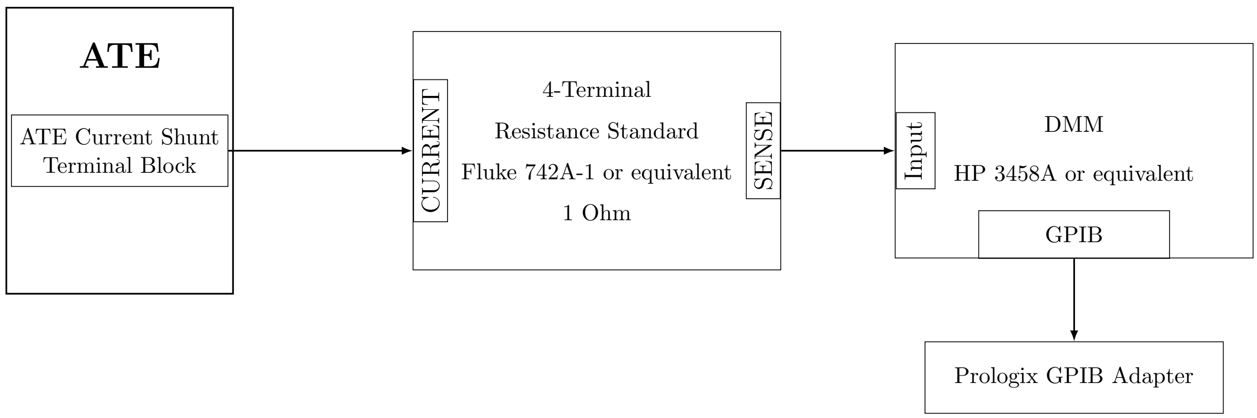


Figure 2.3: ATE → DMM Current Loop

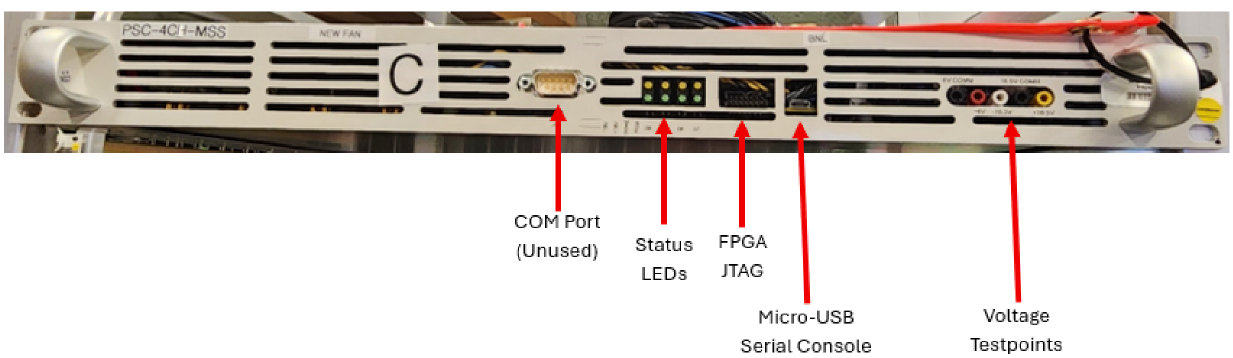
Chapter 3

zPSC Chassis Overview

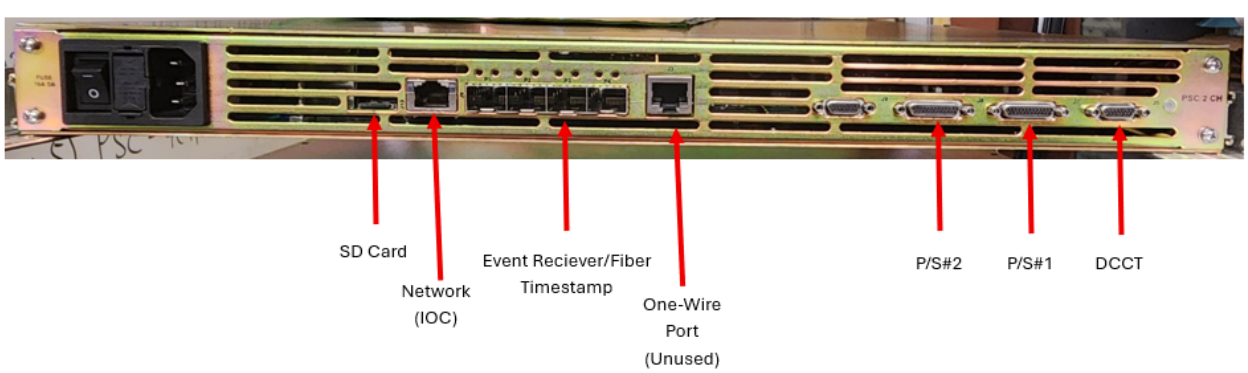
3.1 Introduction

3.2 Front Panels (2- and 4-Channel PSCs)

Both 2- and 4-Channel units share identical front panels:



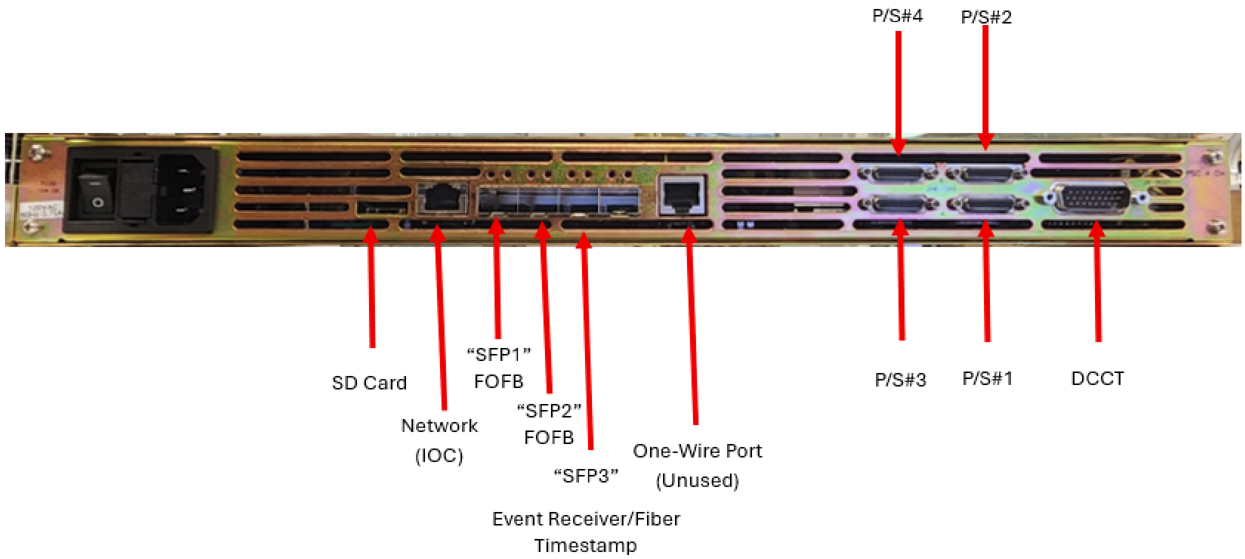
3.3 Rear Panel - 2 Channel



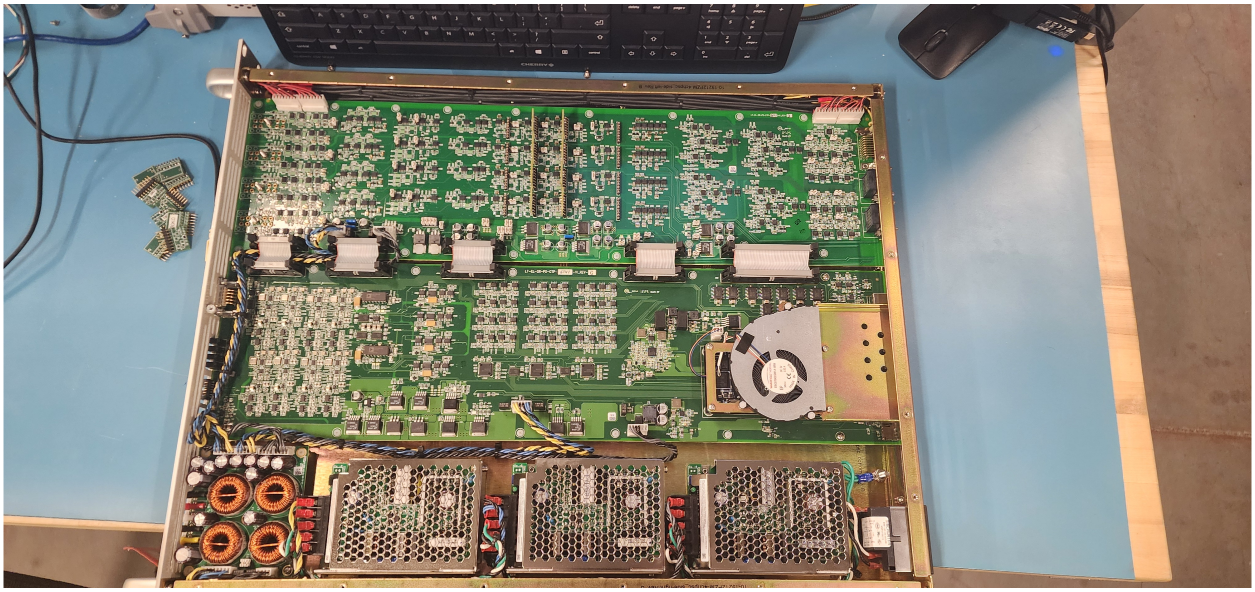
3.4 Internal Component Layout - 2 Channel

3.4.1 Analog Board Components

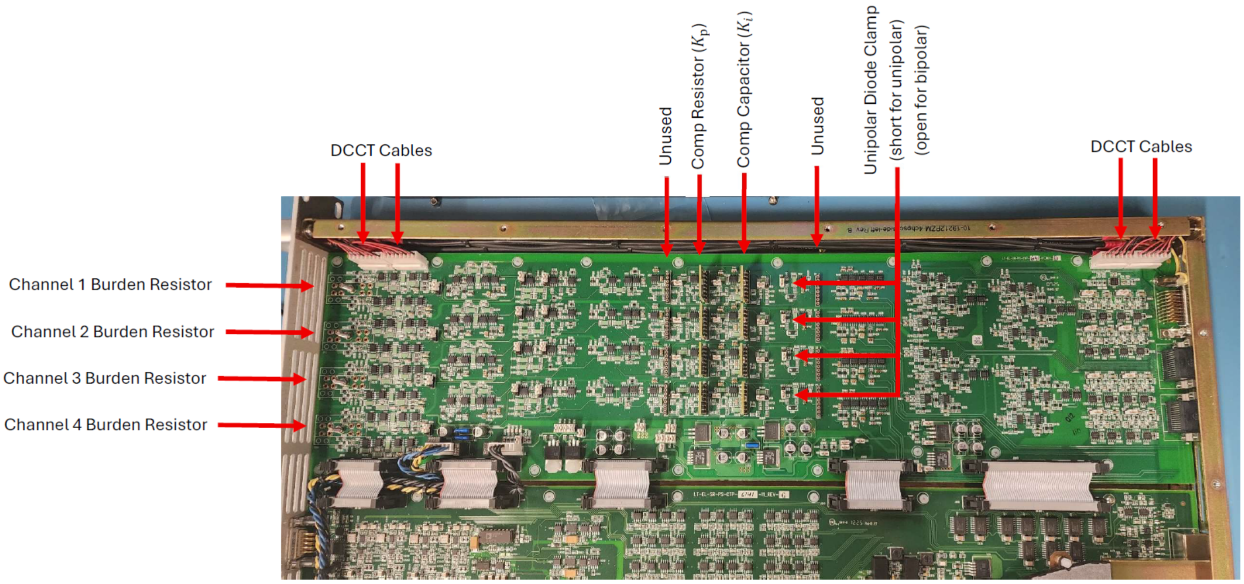
3.5 Rear Panel - 4 Channel



3.6 Internal Component Layout - 4 Channel



3.6.1 Analog Board Components



Chapter 4

EPICS Base and IOCs

4.1 Introduction

There are two IOCs associated with the calibration and test of the zPSC: The ATE IOC, and the zPSC IOC.

4.2 EPICS Installation Overview

EPICS installations performed by M. Capotosto loosely follow the below directory structure:

- `EPICS_BASE=/epics/base` or `/epics/base_[version]` or similar
- `SUPPORT=/epics/support` – or sometimes MODULES instead of SUPPORT is used.

  Modules required for the ATE and PSC are installed here:
  - seq: SNL-Sequencer
  - pscdrv: Portable Streaming Controller Driver
  - asyn: EPICS module for driver and device support
- `epics/iocs` - each IOC is cloned into this repository
- `epics/systemd-softioc` - This is an NSLS-II utility for managing IOCs as a system service

4.3 Installing EPICS Base and Required Modules

1. Create the EPICS user group, and the empty EPICS directory:
   ```
   sudo groupadd epics
   sudo usermod -aG epics $USER
   newgrp epics
   sudo mkdir -p /epics
   ```

```
sudo chown $USER:epics /epics
sudo chmod 2775 /epics
```

2. EPICS Base is cloned into `/epics/base` via:
```
git clone --recursive -b 7.0
   https://github.com/epics-base/epics-base.git /epics/base
cd /epics/base
```

3. Build EPICS:
```
make clean
make
```

4. Create the remaining directory structure:
```
mkdir -p /epics/iocs /epics/modules
```

5. Clone asyn:
   (a) `cd /epics/modules`
   (b) `git clone https://github.com/epics-modules/asyn.git asyn`
   (c) `cd /epics/modules/asyn`
   (d) Set up required files in `configure/RELEASE, configure/RELEASE.LOCAL`
   (e) Build with `make clean`, and `make`

6. Clone Sequencer:
   (a) `cd /epics/modules`
   (b) `git clone https://github.com/epics-modules/sequencer.git seq`
   (c) `cd /epics/modules/seq`
   (d) Set up required files in `configure/RELEASE, configure/RELEASE.LOCAL`
   (e) Build with `make clean`, and `make`

7. Clone PSC Driver:
   (a) `cd /epics/modules`
   (b) `git clone https://github.com/mdavidsaver/pscdrv.git pscdrv`
   (c) `cd /epics/modules/pscdrv`
   (d) Set up required files in `configure/RELEASE, configure/RELEASE.LOCAL`
   (e) Build with `make clean`, and `make`

8. Clone Manage IOCs Utility:

   (a) `cd /epics`
   (b) `git clone https://github.com/NSLS2/systemd-softioc.git`
   (c) Navigate to the repo directory `cd systemd-softioc`
   (d) Make the install script executable and run:
   ```
   sudo chmod +x install.sh
   sudo ./install.sh
   sudo usermod -aG epics softioc
   newgrp
   sudo chown softioc:epics /epics/iocs
   sudo chmod 2775 /epics/iocs
   ```

4.4 Installing the ATE IOC

1. Change to IOC directory `cd /epics/iocs`
2. Clone the repo:
   `git clone https://github.com/capotostobnl/ALSu-PSC_ATE-ioc.git PSCtest`
3. `cd PSCtest`
4. `git switch Released,-UPC/BPC-Split-IOC-Ver-2`
5. Create the `RELEASE.local` file, `touch configure/RELEASE.local`
6. `echo "EPICS_BASE=/epics/base ASYN=/epics/modules/asyn" > /epics/iocs/PSCtest/configure/RELEASE.local`
7. Build the IOC: `make clean`, `make`
8. Update the st.cmd file: `cd iocBoot/iocPSCtest`
   (a) Edit the st.cmd file: `nano st.cmd`
   (b) Update the EPICS_BASE variable if needed.
   (c) Update the IPPORT to match the ATE. Repository has 10.69.26.4:5000 as default. Changing this later from the Phoebus/CSS panels may not work!
9. Find the next available port: `manage-iocs nextport`
10. Configure this in the config file: `nano config`
11. Set the `NAME=PSCTest`, `PORT=` your next port
12. Update the directory path in the root-level st.cmd to point to the right binaries
13. Enroll in manage-iocs as a service: `sudo manage-iocs install PSCtest`

14. Enable auto-start on boot: `sudo manage-iocs enable PSCtest`
15. Check the status, it should be running! `manage-iocs status`
16. And if you want other information on the ioc: `manage-iocs report`